Circle Estimation, Denoising, and Robustness to Shape Misspecification

ST5201 Project Report

National University of Singapore

Wu Xinyu, A0352188X

Wang Xingyi, A0351875U

Lin Yushan, A0352622L

11 November 2026

Part A Circle Estimation

The circle centre and radius are estimated from the first two moments of the noisy observations (see also Al-Sharadqah & Chernov, 2009; Chernov & Lesort, 2005). Subtracting the known noise contribution from the second central moment yields the radius estimate. Under the stated model (National University of Singapore, 2026),

$$X_i = c + RU_i, \qquad Y_i = X_i + \varepsilon_i, \qquad U_i = (\cos\Theta_i, \sin\Theta_i)^\top, \tag{1.1}$$

$$\Theta_i \overset{\text{iid}}{\sim} \text{Uniform}(0, 2\pi), \qquad \varepsilon_i \overset{\text{iid}}{\sim} N_2(0, \sigma^2 I_2), \qquad (\Theta_i)_{i=1}^n \perp (\varepsilon_i)_{i=1}^n. \tag{1.2}$$

Assume $c \in \mathbb{R}^2$, $R > 0$, $n \geq 2$ and known σ. The notation $\|\cdot\|$ denotes the Euclidean norm.

A1 Moment identities and estimators

Since $E(U_i) = 0$, $\| U_i \|^2 = 1$ and $E(\varepsilon_i) = 0$,

$$E(Y_i) = c + R\,E(U_i) + E(\varepsilon_i) = c. \tag{1,3}$$

Moreover, $E\left(U_i^\top \varepsilon_i\right) = 0$ and $E \| \varepsilon_i \|^2 = 2\sigma^2$, so

$$E \| Y_i - c \|^2 = R^2 E \| U_i \|^2 + 2R\,E\left(U_i^\top \varepsilon_i\right) + E \| \varepsilon_i \|^2 = R^2 + 2\sigma^2. \tag{1.4}$$

These identities motivate

$$\hat{c} = \overline{Y} = \frac{1}{n}\sum_{i=1}^n Y_i\,, \qquad \hat{Q} = \frac{1}{n-1}\sum_{i=1}^n \| Y_i - \overline{Y} \|^2 - 2\sigma^2, \; \widehat{R} = \sqrt{\max\{\hat{Q}, 0\}}, \tag{1.5}$$

where $\hat{Q}$ estimates R^2.

A2 Unbiasedness and truncation

By linearity, $E(\hat{c}) = c$. Let $V = R^2 + 2\sigma^2$. Since

$$E(Y_i - c) = 0, \qquad E\left[(Y_i - c)^\top \left(Y_j - c\right)\right] = 0 \quad (i \neq j), \tag{1.6}$$

$$E \| \overline{Y} - c \|^2 = \frac{1}{n^2}\sum_{i=1}^n E \| Y_i - c \|^2 = \frac{V}{n}. \tag{1.7}$$

The sample centring identity gives

$$\sum_{i=1}^n \| Y_i - \overline{Y} \|^2 = \sum_{i=1}^n \| Y_i - c \|^2 - n \| \overline{Y} - c \|^2. \tag{1.8}$$

$$E\left[\sum_{i=1}^n \| Y_i - \overline{Y} \|^2\right] = nV - n(V/n) = (n-1)V, \tag{1.9}$$

$$E\left(\hat{Q}\right) = \frac{(n-1)V}{n-1} - 2\sigma^2 = R^2. \tag{1.10}$$

Thus both $\hat{c}$ and $\hat{Q}$ are unbiased. The denominator $n - 1$ corrects for estimating the centre from the same sample. Using n instead gives expectation $(1 - 1/n)V$ for the sample second central moment, introducing downward bias before noise correction.

Although unbiased, $\hat{Q}$ can be negative when the sample second central moment is below $2\sigma^2$. Truncation ensures a real, nonnegative radius. For $\hat{Q}_+ = \max\{\hat{Q}, 0\}$,

$$E\left(\hat{Q}_+\right) = R^2 + E\left[-\hat{Q}\,\mathbf{1}_{\{\hat{Q}<0\}}\right] \geq R^2, \tag{1.11}$$

The inequality is strict if $\Pr\left(\hat{Q} < 0\right) > 0$. Thus truncation introduces upward bias in the squared-radius estimator. Taking the square root also changes the expectation. By Jensen's inequality,

$$E\left(\hat{R}\right) = E\sqrt{\hat{Q}_+} \leq \sqrt{E\left(\hat{Q}_+\right)}. \tag{1.12}$$

The lower bound $E\left(\hat{Q}_+\right) \geq R^2$ leaves the sign of radius bias undetermined. If truncation is negligible, concavity of the square root tends to induce downward bias. With substantial truncation, the bias depends on both effects.

For small n or low R/σ, negative $\hat{Q}$ can occur more frequently. Truncation then puts a point mass at $\hat{R} = 0$ and can have a substantial effect on bias and variability.

A3 Consistency and the naïve radius

The required moments are finite. By the strong law of large numbers,

$$\overline{Y} \xrightarrow{\text{a.s.}} c, \qquad \frac{1}{n}\sum_{i=1}^{n} \| Y_i - c \|^2 \xrightarrow{\text{a.s.}} V. \tag{1.13}$$

Using the sample centring identity,

$$\frac{1}{n-1}\sum_{i=1}^{n} \| Y_i - \overline{Y} \|^2 = \frac{n}{n-1}\left[\frac{1}{n}\sum_{i=1}^{n} \| Y_i - c \|^2 - \| \overline{Y} - c \|^2\right] \xrightarrow{\text{a.s.}} V. \tag{14}$$

By continuity of truncation and the square root,

$$\hat{c} \xrightarrow{\text{a.s.}} c, \qquad \hat{Q} \xrightarrow{\text{a.s.}} R^2, \qquad \hat{R} \xrightarrow{\text{a.s.}} R. \tag{1.15}$$

Thus all three estimators are consistent. Since $R^2 > 0$, $P(\hat{Q} < 0) \to 0$, so truncation becomes asymptotically irrelevant.

Without noise correction, the naïve estimator satisfies

$$\tilde{R} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} \| Y_i - \overline{Y} \|^2} \xrightarrow{\text{a.s.}} \sqrt{R^2 + 2\sigma^2}. \tag{1.16}$$

For $\sigma > 0$, this limit exceeds R. Increasing n therefore cannot eliminate the naïve estimator's noise-induced inflation, whereas subtracting $2\sigma^2$ yields a consistent radius estimator.

Part B Denoising Methods

B1 Projection onto a circle

For an observation $y \neq c$ and a circle with centre c and radius $R > 0$, the projection is

$$\delta_{\text{proj}}(y; c, R) = c + R\,\frac{y - c}{\|y - c\|}. \tag{2.1}$$

The oracle rule uses the true parameters (c, R). The fitted rule uses $(\hat{c}, \hat{R})$ estimated from the noisy training sample in Part A. Projection preserves the direction from the supplied centre and discards the observed radial distance, replacing it with the supplied radius (Ahn et al., 2001). Angular errors can therefore persist even when the output lies on the true circle.

When $y = c$ and $R > 0$, the direction is undefined, so the implementation returns the fixed point $c + R(1,0)^{\top}$. If the fitted radius is zero, every observation is mapped to $\hat{c}$.

B2 Bayes posterior mean

For a clean point $x(\theta) = c + R(\cos\theta, \sin\theta)$, the uniform angular prior and Gaussian observation model give posterior weights proportional to $\exp\{-\| y - x(\theta) \|^2/(2\sigma^2)\}$ (cf. Banerjee et al., 2005). We approximate the angular integrals on $K = 512$ equally spaced points, normalize the weights, and average the corresponding circle points. For numerical stability, we subtract the largest log weight before exponentiation. On a representative 25-query debugging subset, increasing K to 1024 changed the posterior means by at most 8.90×10^{-16} in Euclidean norm; this is a quadrature check, not a formal performance result.

$$\begin{aligned} &\theta_k = 2\pi k/K, \qquad x_k = c + R(\cos\theta_k, \sin\theta_k), \\ &w_k(y) = \exp\{\ell_k(y) - \max_j \ell_j(y)\}, \qquad \delta_{B,K}(y) = \frac{\sum_k w_k(y) x_k}{\sum_k w_k(y)}. \end{aligned} \tag{2.2}$$

The oracle estimator uses the true c and R; the plug-in estimator substitutes the training-sample estimates $\hat{c}$ and $\hat{R}$ from Part A, so it is not automatically the oracle Bayes rule. Under squared Euclidean loss, $E(\| a - X \|^2 | Y) = \| a - E(X | Y) \|^2 + E(\| X - E(X | Y) \|^2 | Y)$, so the posterior mean minimizes conditional expected loss (see also Efron, 2011). Its weighted average need not lie exactly on the circle: posterior uncertainty across angles can pull the mean inward, which is compatible with pointwise optimality.

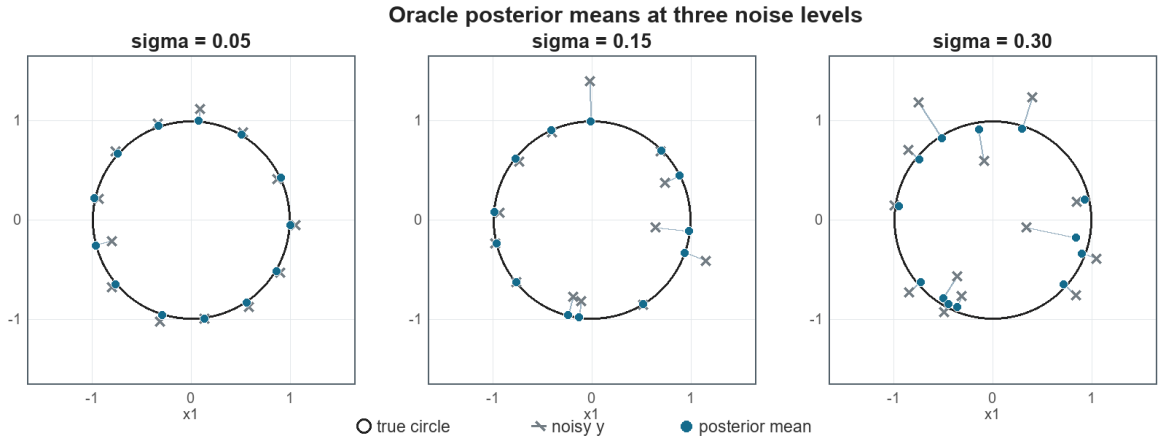


Figure B2. Oracle posterior means for 13 fixed clean angular locations with independent noise at each σ. The panels use the debugging circle $c = (0,0)$, $R = 1$ and illustrate the rule; they are not formal performance results.

B3 Manifold fitting

We apply the two-step weighted contraction map of Yao et al. (2023, arXiv:2304.07680) to the noisy training sample and independent noisy test queries (see also Fefferman et al., 2018; Hein & Maier, 2007; Yao & Xia, 2025). In Step 1, a weighted spherical neighbourhood of radius r_0 produces a local average $F(z)$; the direction from z to $F(z)$ estimates the local direction of contraction. In Step 2, a weighted cylinder aligned with that direction averages nearby training points, retaining local shape information without fitting a global circle. The reference function is manifold_fit_weighted in mf_weighted.py; its fallback diagnostics are returned directly.

The assignment settings are $k = 3$, $r_0 = r_1 = 5\sigma/\log_{10}(n)$, and $r_2 = 10\sigma\sqrt{\log(1/\sigma)}/\log_{10}(n)$, with the natural logarithm in the square root. Here r_1 is the orthogonal cylinder radius and r_2 is its axial half-length. If the first neighbourhood is empty the query z is returned; a numerically zero direction or an empty second-stage cylinder returns $F(z)$. In the formal Part C run, 1,438 of 225,000 MF queries used the empty-sphere fallback (0.6391%); zero queries used either other fallback. MF estimates neither c nor R.

Part C Common Simulation Study

C1 Parameter recovery and pointwise denoising

We evaluated the six estimators introduced in Parts A and B: the raw noisy observation, oracle circle projection, fitted-circle projection, oracle posterior mean, plug-in posterior mean, and weighted manifold fitting (MF). The data-generating circle had centre $c = (0,0)^{\mathsf{T}}$ and radius $R = 1$. For each $(n, \sigma) \in \{100,500,2000\} \times \{0.05,0.15,0.30\}$, 50 independent training/test repetitions were generated after initializing the random-number generator once with seed 5201; every test set contained $m = 500$ points. All six methods within a repetition used the same noisy data, so their comparisons are paired (see Burton et al., 2006, for simulation design).

The fitted circle was obtained from the training sample only. Its parameter-recovery criteria were

$$\| \hat{c} - c \|^2, \qquad \hat{R} - R, \qquad (\hat{R} - R)^2, \tag{3.1}$$

together with the frequency of a negative pre-truncation radius-squared quantity. Denoising performance was measured by

$$\mathrm{MSE}_{\mathrm{den}} = \frac{1}{m}\sum_{i=1}^{m} \| \hat{X}_i - X_i \|^2. \tag{3.2}$$

Table C1 shows that parameter recovery improved strongly with training size. At $\sigma = 0.15$, for example, centre MSE fell from 0.01288 at $n = 100$ to 0.000461 at $n = 2000$. Radius MSE also decreased from 2.74×10^{-4} to 1.17×10^{-5}. The radius bias was small and changed sign across settings, whereas the negative-quantity count was zero in all 450 fitted circles. Thus the main finite-sample issue was variability rather than a persistent directional bias or numerical breakdown.

Noise dominated denoising error. At $n = 2000$, raw MSE increased from 0.00497 at $\sigma = 0.05$ to 0.18175 at $\sigma = 0.30$. Every denoising method improved substantially on the raw data. At low noise, the two oracle methods were essentially tied (MSE 0.00250), while the fitted and plug-in methods were only slightly worse (approximately 0.00279). At $\sigma = 0.30$, the oracle posterior mean and plug-in posterior mean had MSE 0.09560 and 0.09602, respectively, compared with 0.09851 for oracle projection, 0.09902 for fitted projection, and 0.10651 for MF. Hence posterior averaging can outperform exact projection under squared-error loss: the posterior mean is not required to lie on the circle, and shrinkage can reduce pointwise risk.

Table C1. Circle-parameter recovery (50 repetitions per setting).

n	σ	**Centre MSE** Mean (MCSE)	**Radius bias** Mean (MCSE)	**Radius MSE** Mean (MCSE)
100	0.05	0.008956 (.000978)	-0.000208(.000871)	0.0000372(.00000739)
100	0.15	0.012883(.00189)	-0.006165(.00220)	0.0002742(.0000758)
100	0.30	0.013131(.00154)	0.006083(.00465)	0.0010973(.000214)
500	0.05	0.002362 (.000352)	-0.000750(.000327)	0.00000579(.00000135)
500	0.15	0.001811(.000237)	0.000661(.000926)	0.0000424(.0000104)
500	0.30	0.001826 (.000242)	-0.002159(.00208)	0.0002172(.0000383)
2000	0.05	0.000581(.000103)	-0.000047(.000173)	0.00000146(.000000283)
2000	0.15	0.000461(.0000625)	-0.000092(.000489)	0.0000117(.0000019)
2000	0.30	0.000698 (.0000915)	0.000646(.00111)	0.0000605(.0000124)

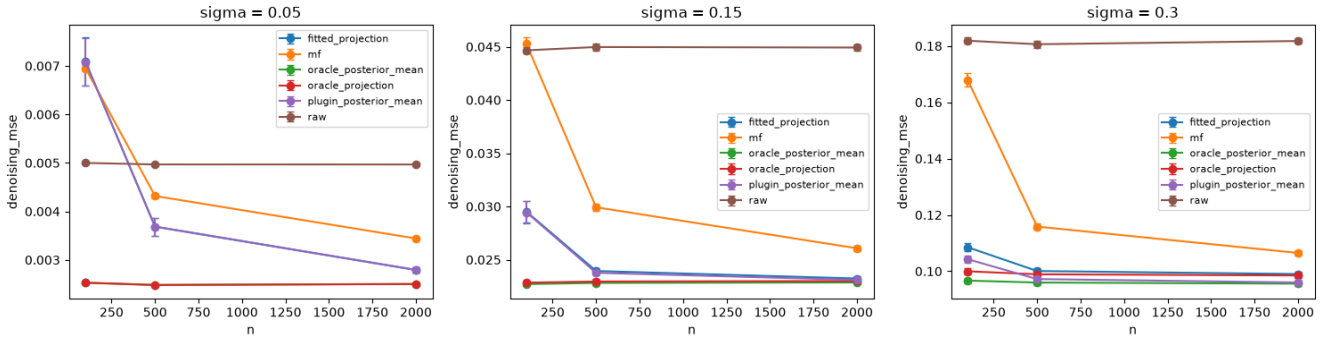


Figure C1. Denoising performance across training sizes and noise levels.

C2 Geometric recovery

Denoising MSE does not completely describe recovery of the underlying manifold (see also Genovese et al., 2012). We therefore used two complementary geometric criteria. For the unit circle,

$$E_{\mathrm{rad}} = \frac{1}{m}\sum_{i=1}^{m}(\| \hat{X}_i - c \| - R)^2 \tag{3.3}$$

measures the average squared radial departure of the estimated points from the true circle. Coverage error instead measures whether the estimated point cloud represents the full circle (cf. Fan et al., 2017):

$$E_{\mathrm{cov}} = \frac{1}{L}\sum_{\ell=1}^{L} \min_{1\le i\le m} \| g_\ell - \hat{X}_i \|^2, \qquad L = 720, \tag{3.4}$$

where g_ℓ are equally spaced points on the true circle. The clean test sample has non-zero coverage error because only 500 random angles are observed; in the formal runs its baseline was about 8×10^{-5}. This clean-test value is a spacing reference, not a strict lower bound.

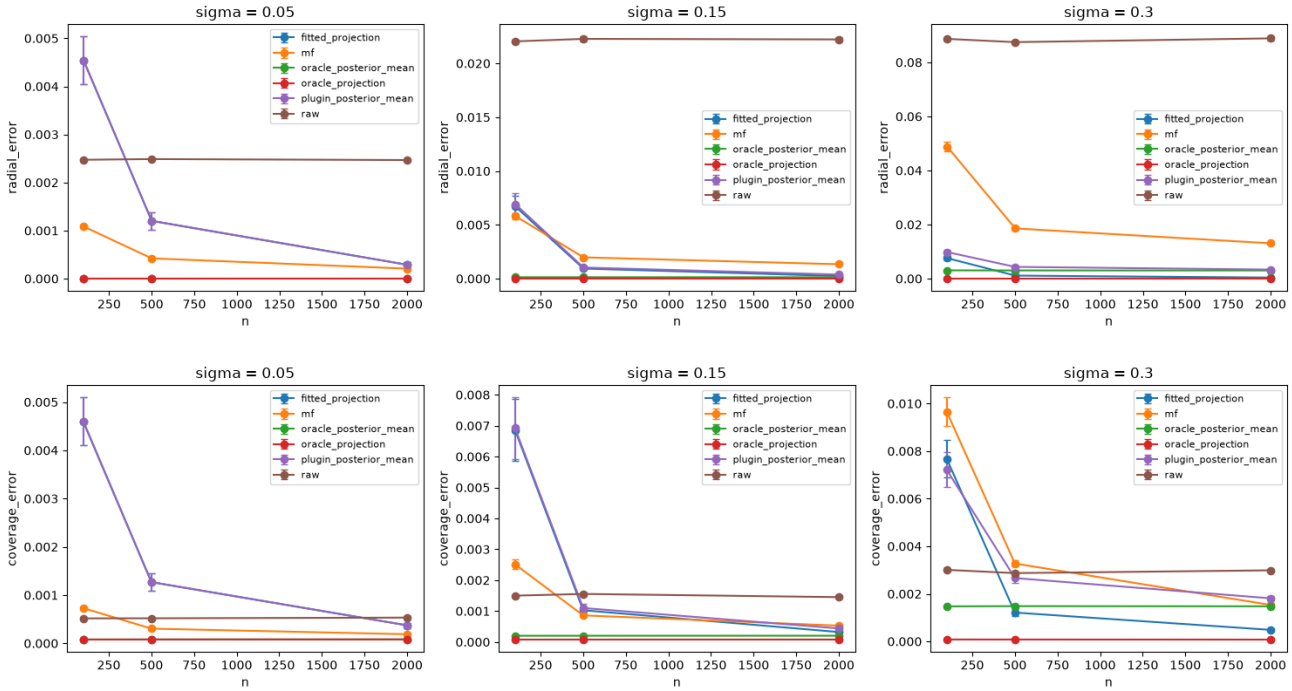


Figure C2. Radial and coverage errors by n and σ (means over 50 repetitions).

The two criteria produce different rankings because they test different geometric properties. Oracle projection has radial error at numerical zero by construction and near-baseline coverage. At $n = 2000, \sigma = 0.15$, fitted projection attained radial error 2.44×10^{-4} and coverage error 3.22×10^{-4}, while MF attained a larger radial error (1.34×10^{-3}) but a comparable-order coverage error (5.29×10^{-4}). Raw observations had much larger radial error (0.02223) yet coverage error of only 0.00145: a dispersed cloud may place points near many angular locations while remaining far from the circle. Conversely, the posterior mean can achieve low denoising MSE through shrinkage while having non-zero radial error. Radial fidelity, angular coverage, and pointwise squared loss should therefore be reported separately.

Table C2. Geometric recovery at $n = 2000$, $\sigma = 0.15$ (mean [MCSE], 50 repetitions).

Method	**Radial error**	**Coverage error**
Raw observation	2.223e-02 (1.86e-04)	1.452e-03 (2.28e-05)
Oracle projection	4.831e-33 (6.04e-35)	7.916e-05 (1.49e-06)
Fitted projection	2.442e-04 (3.15e-05)	3.220e-04 (3.17e-05)
Oracle posterior mean	1.346e-04 (3.28e-07)	2.036e-04 (1.52e-06)
Plug-in posterior mean	3.823e-04 (3.61e-05)	4.388e-04 (3.44e-05)
Manifold fitting (MF)	1.341e-03 (4.09e-05)	5.293e-04 (1.08e-05)
Clean test reference	—	$\approx 8 \times 10^{-5}$

C3 Evidence and interpretation

Figure C3 gives an equal-axis visualization of one prespecified formal repetition. It makes the preceding distinction visible: projection methods concentrate points on an estimated curve; posterior means can contract toward the centre; and MF follows local data geometry with some residual spread. This figure is illustrative only—the numerical conclusions above use all 50 repetitions in every setting.

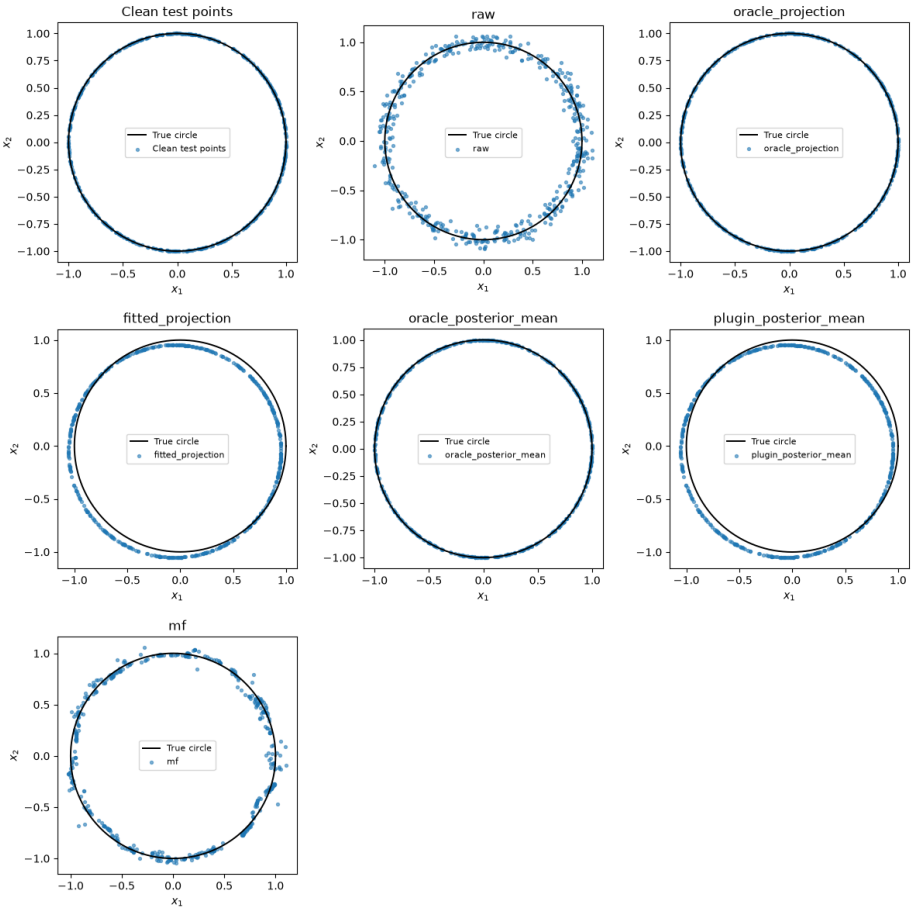


Figure C3. Representative outputs of the six denoising rules.

Increasing n principally improved methods that estimate global circle parameters. At $\sigma = 0.15$, fitted-projection denoising MSE decreased from 0.02951 at $n = 100$ to 0.02324 at $n = 2000$, and its coverage error decreased from 0.00686 to 0.000322. The oracle methods, whose circle is already known, changed much less with n. Increasing σ raised denoising error for every method and made posterior contraction more consequential. It also exposed a distinction between exact geometric adherence and squared-error optimality: at $n = 2000, \sigma = 0.30$, oracle projection remained exactly on the circle, but oracle posterior mean had the lower denoising MSE.

For each reported mean $\bar{Z}$, simulation uncertainty was summarized by

$$\mathrm{MCSE}(\bar{Z}) = s_Z/\sqrt{50}, \tag{3.5}$$

using the repetition-level sample standard deviation s_Z (Morris et al., 2019). For example, the fitted-projection denoising mean at $(n, \sigma) = (100, 0.15)$ was 0.02951 with MCSE 0.00101, whereas at $(2000, 0.15)$ it was 0.02324 with MCSE 0.000218. MCSE quantifies numerical uncertainty of the Monte Carlo mean, not uncertainty for a new observation; close method rankings should therefore be interpreted alongside their MCSEs and paired design rather than from point estimates alone.

The MF diagnostic log recorded 1,438 empty-sphere fallbacks among 225,000 queries (0.6391%), with no zero-direction or empty-cylinder fallbacks. These events were concentrated in the smallest-n, lowest-noise setting and

did not invalidate the run, but they document the finite-sample behaviour of the local estimator. Overall, there is no universally best estimator. Oracle projection is optimal for radial adherence; oracle/posterior methods are competitive for denoising; fitted and plug-in methods approach their oracle counterparts as parameter recovery improves; and MF can preserve local support well but is more variable in sparse settings. Good parameter recovery generally benefits fitted-circle methods, but it does not force identical rankings under denoising, radial, and coverage loss.

Part D Recovery under Ellipse Misspecification

D1 Research question and conjectures

Part D examines robustness to model misspecification. The true latent curve was changed from a circle to the ellipse

$$X(\theta) = c + (\cos\theta,\, b\sin\theta)^{\top}, \qquad \theta \sim \mathrm{Unif}(0,2\pi), \tag{4.1}$$

with $c = (0,0)^{\top}$ and $b \in \{1,0.9,0.75\}$. We asked how the raw data, fitted-circle projection, plug-in circle posterior mean, and MF behave as the true shape departs from a circle. Before observing the results, we expected the circle-based methods to develop a systematic approximation error as b decreased; MF, which uses local geometry rather than a global circle equation, was expected to be more adaptive when training data were sufficiently dense. We also expected increasing n to reduce estimation variability but not eliminate the structural bias of fitting a circle to a non-circular ellipse (cf. Fitzgibbon et al., 1999; Kanatani, 1994). These statements are conjectures; the conclusions below are based on the formal simulation.

D2 Experimental design and evaluation

The formal experiment fixed $\sigma = 0.15$, used $n \in \{100,500,2000\}$, $m = 500$ test points, 50 repetitions per (b, n) setting, and the same seed discipline as Part C. Four methods were compared: raw, fitted-circle projection, plug-in circle posterior mean, and MF. The circle posterior remained intentionally misspecified; it was not an oracle ellipse posterior.

Because the target was an ellipse, Part C's circle radial error was inappropriate. Let G_b be a dense 4096-point grid on the true ellipse. We used the two directional squared nearest-neighbour errors (cf. Fan et al., 2017)

$$d_{\mathrm{out}\to\mathrm{ell}} = \frac{1}{m}\sum_{i=1}^{m} \min_{g\in G_b} \| \hat{X}_i - g \|^2, \qquad d_{\mathrm{ell}\to\mathrm{out}} = \frac{1}{|G_b|}\sum_{g\in G_b} \min_{1\le i\le m} \| g - \hat{X}_i \|^2. \tag{4.2}$$

The first measures whether outputs lie near the ellipse; the second measures whether the output cloud covers the ellipse. Denoising MSE was computed against the clean latent point paired with each noisy observation. All summary means and MCSEs were calculated from the 50 repetition-level values.

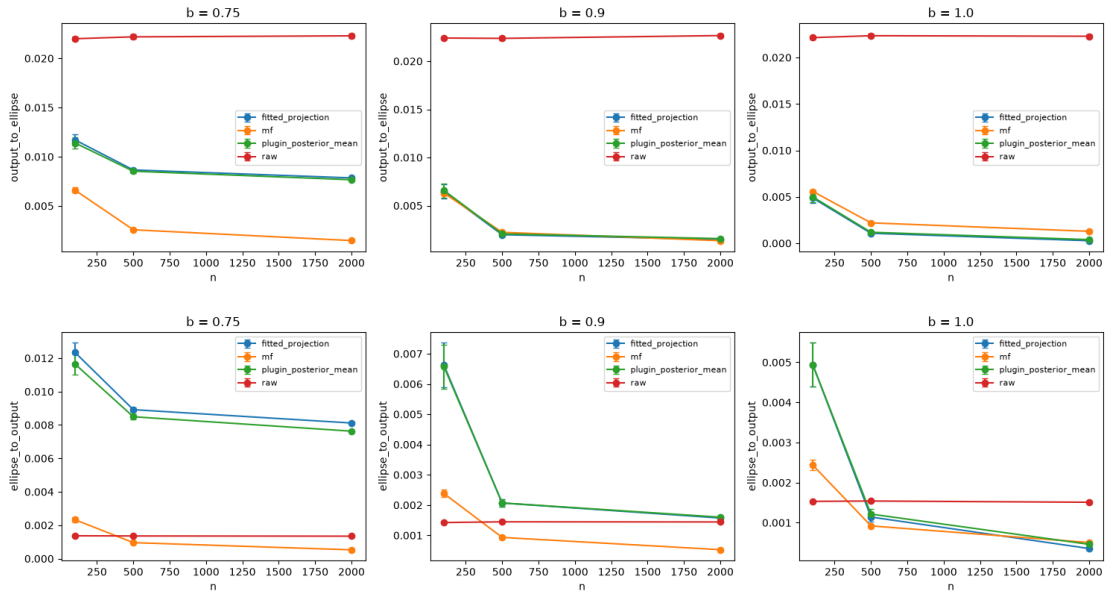


Figure D1. Directed geometric errors under increasing ellipse misspecification.

D3 Results

Table D1 focuses on $n = 2000$, where estimation noise is relatively small. When $b = 1$, the circle methods were correctly specified and had the lowest denoising MSE (about 0.0232), while MF had MSE 0.02637. As the ellipse became flatter, the circle-based geometric errors increased sharply. For fitted projection, output-to-ellipse error rose from 0.000355 at $b = 1$ to 0.008117 at $b = 0.75$; plug-in posterior mean showed the same pattern. MF was comparatively stable: at $b = 0.75$, its output-to-ellipse and ellipse-to-output errors were 0.000527 and 0.001477, respectively, and its denoising MSE (0.02618) was lower than those of fitted projection (0.03150) and plug-in posterior mean (0.03092).

Table D1. Part D means at $\mathbf{n = 2000}$ (50 repetitions; $\boldsymbol{\sigma = 0.15}$).

b	Method	Denoising MSE	Output → ellipse	Ellipse → output
1.00	raw	0.04501	0.001510	0.022288
1.00	fitted projection	0.02329	0.000355	0.000271
1.00	plug-in posterior	0.02314	0.000459	0.000390
1.00	MF	0.02637	0.000505	0.001288
0.90	raw	0.04531	0.001442	0.022658
0.90	fitted projection	0.02451	0.001572	0.001493
0.90	plug-in posterior	0.02429	0.001600	0.001578
0.90	MF	0.02616	0.000525	0.001359
0.75	raw	0.04503	0.001344	0.022292
0.75	fitted projection	0.03150	0.008117	0.007846
0.75	plug-in posterior	0.03092	0.007629	0.007659
0.75	MF	0.02618	0.000527	0.001477

The sample-size comparison supports the distinction between variance and misspecification. At $b = 0.75$, MF denoising MSE decreased from 0.04615 at $n = 100$ to 0.03018 at $n = 500$ and 0.02618 at $n = 2000$. Its output-to-ellipse error decreased from 0.00659 to 0.00257 and then 0.000527. The fitted-circle method improved more modestly: its output-to-ellipse error decreased from 0.01170 at $n = 100$ to 0.00812 at $n = 2000$, leaving a

substantial approximation error. More data therefore helped all estimated methods, but only the locally adaptive MF continued to approach the non-circular target closely.

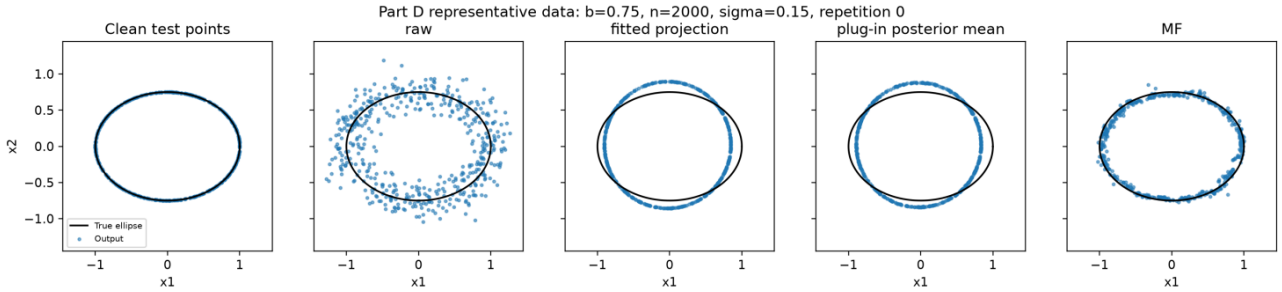


Figure D2. Representative misspecified-ellipse outputs (repetition 0).

Figure D2 visually confirms the aggregate results. The fitted projection and plug-in posterior mean remain approximately circular, creating systematic gaps relative to the upper and lower portions of the ellipse. MF follows the elliptical support more closely, although it retains local scatter. The raw cloud has low output-to-ellipse error because some noisy points remain close to the curve, but its ellipse-to-output error and denoising MSE are large. This is another ranking reversal: one-sided proximity alone does not imply either good coverage or accurate pointwise denoising.

D4 Discussion and limitations

The experiment supports the pre-specified conjecture that a global circle model incurs structural error when the shape is only approximately circular. At mild misspecification ($b = 0.9$), the circle methods remained competitive for denoising, whereas MF gave the smallest output-to-ellipse error. Under stronger misspecification ($b = 0.75$) and large n, MF was best on all three reported criteria among the four evaluated methods. Increasing n improved estimation and local neighbourhood quality, but it did not remove the geometric bias of the fitted circle.

These conclusions are limited to centred axis-aligned ellipses, uniform angular sampling, $\sigma = 0.15$, the specified tuning rules, and the three tested aspect ratios. The experiment did not include a correctly specified ellipse posterior, so it compares practical circle-based procedures with MF rather than establishing an ellipse-oracle benchmark. The two directed grid distances are finite-grid approximations and emphasize different failure modes. Within this design, however, the answer to the research question is clear: circle-based methods remain useful for small departures from circularity, but their global misspecification becomes dominant as eccentricity increases; MF is more robust when sufficient training data are available.

Final Reflection

No estimator was uniformly best because the project considered distinct targets. Projection was unmatched for exact radial adherence under the correct circle model, whereas posterior averaging could reduce pointwise squared error through shrinkage. Fitted and plug-in procedures improved markedly with training size because circle-parameter uncertainty decreased; oracle methods were comparatively insensitive to n. Higher noise increased every denoising error and made rankings under squared loss diverge from rankings under geometric loss. Under ellipse misspecification, increasing n reduced variance but could not eliminate the structural bias of a global circle model, while MF benefited from denser local neighbourhoods. Apparent disagreements between methods should therefore be investigated using paired repetition-level differences, MCSEs, both directions of geometric error, diagnostic fallback counts, and representative equal-axis plots rather than a single summary metric.

Group Contribution Statement

Lin Yushan. Derived and implemented the moment-based circle estimator, implemented the oracle and fitted projection rules, and developed and checked the C1 parameter-recovery metrics.

Wu Xinyu. Implemented and documented the Bayes posterior mean and manifold fitting methods and evaluated pointwise denoising performance for C1.

Wang Xingyi. Integrated the common simulation, led geometric recovery and the cross-metric interpretation, conducted the ellipse investigation, and assembled the group report.

AI Use Disclosure

OpenAI ChatGPT assisted with coding and debugging for circle estimation, projection and parameter-recovery metrics, and with formatting the report. The estimator and projection were checked against hand-computable examples, including the zero-radius case. A seeded simulation checked the parameter-recovery calculations.

References

Ahn, S. J., Rauh, W., & Warnecke, H.-J. (2001). Least-squares orthogonal distances fitting of circle, sphere, ellipse, hyperbola, and parabola. *Pattern Recognition, 34*(12), 2283–2303. https://doi.org/10.1016/S0031-3203(00)00152-7

Al-Sharadqah, A., & Chernov, N. (2009). Error analysis for circle fitting algorithms. *Electronic Journal of Statistics, 3*, 886–911. https://doi.org/10.1214/09-EJS419

Banerjee, A., Dhillon, I. S., Ghosh, J., & Sra, S. (2005). Clustering on the unit hypersphere using von Mises-Fisher distributions. *Journal of Machine Learning Research, 6*, 1345–1382. https://www.jmlr.org/papers/v6/banerjee05a.html

Burton, A., Altman, D. G., Royston, P., & Holder, R. L. (2006). The design of simulation studies in medical statistics. *Statistics in Medicine, 25*(24), 4279–4292. https://doi.org/10.1002/sim.2673

Chernov, N., & Lesort, C. (2005). Least squares fitting of circles. *Journal of Mathematical Imaging and Vision, 23*(3), 239–252. https://doi.org/10.1007/s10851-005-0482-8

Efron, B. (2011). Tweedie's formula and selection bias. *Journal of the American Statistical Association, 106*(496), 1602–1614. https://doi.org/10.1198/jasa.2011.tm11181

Fan, H., Su, H., & Guibas, L. J. (2017). A point set generation network for 3D object reconstruction from a single image. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 2463–2471). IEEE. https://doi.org/10.1109/CVPR.2017.264

Fefferman, C., Ivanov, S., Kurylev, Y., Lassas, M., & Narayanan, H. (2018). Fitting a putative manifold to noisy data. In S. Bubeck, V. Perchet, & P. Rigollet (Eds.), *Proceedings of the 31st Conference on Learning Theory* (Vol. 75, pp. 688–720). PMLR. https://proceedings.mlr.press/v75/fefferman18a.html

Fitzgibbon, A., Pilu, M., & Fisher, R. B. (1999). Direct least square fitting of ellipses. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 21*(5), 476–480. https://doi.org/10.1109/34.765658

Genovese, C. R., Perone-Pacifico, M., Verdinelli, I., & Wasserman, L. (2012). Manifold estimation and singular deconvolution under Hausdorff loss. *The Annals of Statistics, 40*(2), 941–963. https://doi.org/10.1214/12-AOS994

Hein, M., & Maier, M. (2007). Manifold denoising. In B. Schölkopf, J. Platt, & T. Hofmann (Eds.), *Advances in neural information processing systems 19* (pp. 561–568). MIT Press. https://proceedings.neurips.cc/paper_files/paper/2006/hash/a0b83c02d720415dada82e08bc09e9f3-Abstract.html

Kanatani, K. (1994). Statistical bias of conic fitting and renormalization. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 16*(3), 320–326. https://doi.org/10.1109/34.276132

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine, 38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

National University of Singapore. (2026). *Recovering a circle from noisy data* [ST5201 project brief].

Yao, Z., Su, J., Li, B., & Yau, S.-T. (2023). *Manifold fitting* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2304.07680

Yao, Z., & Xia, Y. (2025). Manifold fitting under unbounded noise. *Journal of Machine Learning Research, 26*(45), 1–55. https://www.jmlr.org/papers/v26/21-0039.html